

Literature-Implied Subcase for arXiv:math/0611724

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Status

This is a `literature_implied_answer` (partial subcase / reformulated resolvent condition) packet. It does not claim a new proof, and it does not claim a complete answer to the exact finite-cotype formulation in Haak–van Neerven [1]. It records that a later theorem of Abreu–Haak–van Neerven [2] proves the central invariant-measure equivalence under Pisier property (α) .

Source Conjecture

In the final section of [1], after proving that existence of an invariant measure implies uniform γ -radonification of the sector resolvent family

$$\{\sqrt{\lambda}R(\lambda, A)B : \lambda \in S_{\vartheta}\},$$

Haak–van Neerven formulate the stochastic Weiss conjecture. In their displayed formulation, E has finite cotype, $-A$ is injective and sectorial of angle less than $\pi/2$, and $-A$ admits a bounded H^∞ calculus. The conjecture asks for equivalence of:

1. the stochastic Cauchy problem admits an invariant measure;
2. $(-A)^{-1/2}B$ is γ -radonifying;
3. the sector family $\{\sqrt{\lambda}R(\lambda, A)B\}$ is uniformly γ -radonifying for some/every sector angle $0 < \vartheta < \pi/2$.

The authors note that the implication from (2) to (1) is known, and that the remaining hard direction is from the resolvent condition to $(-A)^{-1/2}B \in \gamma(H, E)$, or directly to invariant measure.

Supporting Theorem

The later paper [2] explicitly cites [1] and states in its abstract that it solves the stochastic Weiss conjecture proposed there. Its main theorem assumes that E has Pisier property (α) and that $-A$ admits a bounded H^∞ calculus of angle less than $\pi/2$. It proves the equivalence of invariant measure, $(-A)^{-1/2}B \in \gamma(H, E)$, a $\gamma(L^2(\mathbb{R}_+, d\lambda/\lambda; H), E)$ resolvent condition, and a dyadic Gaussian resolvent condition.

Idea of the Identification

The source conjecture’s central analytic content is the equivalence between existence of an invariant measure and the half-power condition $(-A)^{-1/2}B \in \gamma(H, E)$. The later theorem proves exactly this equivalence under property (α) , and it supplies a precise stochastic-resolvent condition in place of the original uniform family condition. Property (α) includes Hilbert spaces and L^p spaces, and the supporting paper recalls Pisier’s theorem that a Banach lattice has property (α) if and only if it has finite cotype. Thus the literature theorem answers a substantial finite-cotype lattice/Hilbert subcase of the 2006 programme.

Theorem 1. *Let E have property (α) . Suppose $-A$ is injective, sectorial of angle less than $\pi/2$, has dense range, and admits a bounded H^∞ calculus of angle less than $\pi/2$ on E . Let $B : H \rightarrow E_{-1}$ be bounded. Then the stochastic Cauchy problem admits an invariant measure on E if and only if $(-A)^{-1/2}B \in \gamma(H, E)$.*

Proof. This is the implication between assertions (1) and (2) in Theorem 1.1 of Abreu–Haak–van Neerven [2]. The same theorem also proves equivalence with the supporting paper’s $\gamma(L^2(d\lambda/\lambda))$ and dyadic Gaussian resolvent conditions. $\square$

Limitations

This packet is deliberately scoped. The original displayed conjecture in [1] assumes finite cotype and uses uniform γ -radonification of the whole sector resolvent family. The supporting paper assumes property (α) and formulates the resolvent side through Gaussian square-function/dyadic conditions. Therefore the packet should not be counted as a full answer to the exact finite-cotype uniform-family statement. The apparent remaining territory is the finite-cotype case without property (α) , and the precise relationship between the original uniform-family condition and the later dyadic/Gaussian condition outside the supporting paper’s hypotheses.

Search Evidence

The run indexes were searched for `0611724`, `gamma-radonifying`, `stochastic Weiss`, `Weiss conjecture`, `admissibility`, and `analytic semigroup`. No prior packet or attempt for this target was found. Local source comparison identified arXiv:1206.3656 as the decisive supporting paper. Web/arXiv searches on June 15, 2026 for stochastic Weiss conjecture, finite cotype, property (α) , and gamma-radonifying variants found the original arXiv source and arXiv:1206.3656, but no later primary source removing the property (α) restriction.

Human Review Recommendation

Review as a provenance/status packet. The main points to check are whether a later source has closed the finite-cotype gap, and whether the published version of [2] labels the main theorem differently from the arXiv source. This should not be counted as an original proof or a full solution of the 2006 conjecture as displayed.

References

- [1] B. H. Haak and J. van Neerven, *Uniformly gamma-radonifying families of operators and the stochastic Weiss conjecture*, arXiv:math/0611724.
- [2] J. Abreu, B. Haak, and J. van Neerven, *The stochastic Weiss conjecture for bounded analytic semigroups*, arXiv:1206.3656.